

HIDDE FOKKEMA

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EXPERIENCE

Machine Learning Scientist

Booking.com  2025/07-2025/09

- Visiting Researcher, researching the impact of using various bandit optimisation algorithms for e-commerce applications.

Junior Quantitative Consultant

Amsterdam Data Collective  2018/03-2021/06

- Built a full-stack web application, using Django, Plotly and PyTorch, for HelloFresh leveraging NLP techniques to analyse consumer reviews and optimise the consumer feedback process. Set up CI/CD pipelines to quickly incorporate any client feedback.
- Developed a tool that automatically scrapes, analyses and recommends biomedical articles based, streamlining the process of finding new leads at NLC, a health-tech incubator.
- Designed and implemented a dashboard to visualise flight occupation and pricing data for Accelya, leading to insights into the profitability of several flight routes.
- Developed key components of an Expected Credit Loss model for the consumer loan portfolio at Aegon Bank as part of the IFRS 9 report to the European Central Bank.

Teaching Assistant

University of Amsterdam  2018/05-2024/08

As a student and during the PhD, assisted as a TA in several Bachelor and national Master courses for the mathematics programme:

- **Bachelor:** Was responsible for organizing exercise class, grading exams and developing solution sheets for the courses: *stochastics 2: statistics*, *Bayesian statistics* and *Introduction to numerical mathematics*.
- **Master:** Provided lecture style solution classes and graded exams for master courses *machine learning theory (M1)* and *stochastic intergration (M2)*.

International Volunteer

European Youth Parliament  2014/09-2019/03

The European youth Parliament is an international network active in over 40 countries in Europe. Its goal is to provide young people an interactive educational platform.

- (2017-2019), Treasurer of EYP NL. I oversaw the financial well-being of EYP NL, with yearly budgets of €200.000.
- (2016-2017), Head-organiser of the 17th National Selection Conference of EYP NL in Maastricht. Fundraised €15.000 and oversaw the coordination of a volunteer team of over 50 people responsible for 150 participants.

EDUCATION

PhD in Mathematical Machine Learning

University of Amsterdam  2021/09 - Present

Thesis title: *Mathematical Foundations of Explainable AI and Advances in Bandit Optimisation*.

Supervised by: dr. Tim van Erven.

Submission date: 2025/07/16

Conducted research and developed theory for a wide range of topics in ML, covering attribution methods, counterfactual explanations, concept-based models and bandit convex optimisation

- Proposed and proved several impossibility results for attribution and counterfactual methods.
- Introduced new estimators for concept-based models, which enable the incorporation of techniques from causal representation learning.
- Advanced the best known efficient algorithm for convex bandit optimisation in the stochastic and adversarial case.

Published in JMLR, AISTATS, COLT and NeurIPS.

Reviewed for AISTATS 2024, TMLR, UAI 2025 and Journal of Approximate Reasoning.

MSc. in Mathematics

University of Amsterdam  2019/09 - 2021/08

Stochastics track, GPA: 8.9 (*Cum Laude*).

Thesis title: *Stability and Computation of Martingale Optimal Transport*.

Supervision: prof. dr. Peter Spreij, dr. Sonja Cox.

- Courses in machine learning theory, stochastic processes, stochastic integration, topological data analysis, numerical methods for linear algebra and PDEs and differential geometry.

BSc. in Mathematics

BSc. in Physics and Astronomy

University of Amsterdam  2015/09 - 2019/08

Focused on courses in probability, numerical methods, quantum mechanics and electro dynamics. GPA: 8.1.

Thesis title: *Learning and Thermodynamics*.

Supervision: dr. Bas Kleijn, dr. Greg Stephens

- Extra minor Artificial Intelligence (36 EC) with courses in machine learning, reinforcement learning, NLP and fuzzy logic.

SKILLS

Languages	Dutch (native), English (fluent), German (intermediate), French (basic)
Program Languages	Python, L ^A T _E X, Mathematica, Javascript/Typescript, Matlab, R, HTML, Rust
ML frameworks	Scikit-Learn, PyTorch, JAX
Tools	Vim, Git, Docker, SLURM, CI/CD (Github actions and Gitlab CI/CD)
Databases	SQL, MySQL, DuckDB, Django

PUBLICATIONS

Published

- H. Fokkema, T. van Erven, S. Magliacane, [Sample-efficient Learning of Concepts with Theoretical Guarantees: from Data to Concepts without Interventions](#), To appear at the *Conference on Neural Information Processing Systems*, 2025
- G. König, H. Fokkema, T. Freiesleben, C. Mendler-Dünner, U. von Luxburg, [Performative Validity of Recourse Explanations](#), To appear at the *Conference on Neural Information Processing Systems*, 2025
- H. Fokkema, D. van der Hoeven, T. Lattimore, J. Mayo, [Online Newton Method for Bandit Convex Optimisation](#), *Conference on Learning Theory*, PMLR, vol. 196, 2024.
- H. Fokkema, D. Garreau, T. van Erven, [The Risks of Recourse in Binary Classification](#), *International Conference on AI & Statistics*, PMLR, vol. 238, pp. 550–558, 2024.
- H. Fokkema, R. de Heide, T. van Erven, [Attribution-based Explanations that Provide Recourse Cannot be Robust](#), *Journal of Machine Learning Research*, vol. 24, no. 360, pp. 1-37, 2023.

SELECTED TALKS

- Explainable Learning & Reasoning Workshop at Cardiff University, [Performativity and Risks of Algorithmic Recourse](#), June 2025;
- Tübingen AI Center Talks, Using Causal Representation Learning for to formalize concept extraction from learned representations, November 2024;
- International Conference on Machine Learning, [Attribution-based Explanations that Provide Recourse Cannot be Robust](#), poster, July 2024;
- Conference on Learning Theory, [Online Newton Method for Bandit Convex Optimisation](#), June 2024
- International Theory of Interpretable AI Seminar, [Attribution-based Explanations that Provide Recourse Cannot be Robust](#), May 2024;
- Ministry of Justice and Security, [Some Theoretical Limitations of Explainability Methods](#), March 2024
- Saarland University Workshop on Interpretability and Recourse, [The Risks of Recourse in Binary Classification](#), October 2023;
- Amsterdam Business School, [The Risks of Recourse in Binary Classification](#), September 2023;
- Xomnia Data Seminar, [Some Theoretical Limitations of Explainability Methods](#), September 2023;
- 1st Nice workshop on Interpretability, [Attribution-based Explanations that Provide Recourse Cannot be Robust](#), November 2022;
- 1st Workshop Mathematics & Artificial Intelligence, [Attribution-based Explanations that Provide Recourse Cannot be Robust](#), poster presentation, June 2022;